2.3 What is e-learning?

In this lesson

You will learn:

- how computers are used for e-learning.

What is e-learning?

An important use of computers in education is for **e-learning**. The 'e-' in e-learning stands for electronic. E-learning is like classroom learning in many ways, but it takes place over a local area network or the internet.



With e-learning, you can decide when and where you want to study. E-learning doesn't have a timetable, although there are usually set dates to submit your work. In the classroom, materials may be printed on paper or projected on the whiteboard. In e-learning, all the learning is done on a computer screen.

Who uses e-learning?

E-learning is often used in the workplace for training people. It is used to help people get new skills, for example, IT skills. It is also used when people need to know about new rules or ways of doing things, for example, if a government passes a new law that people in a particular job must know about.

E-learning is never the main type of learning in schools. It may be used by teachers to give you extra help. E-learning can also be used to give you extra challenges. Students can use e-learning at home to do extra study.

Multimedia and e-learning

In your classroom you have books to learn from. Your teacher gives you information. Sometimes, this information is given to the whole class using the board. At other times, you might work in small groups around a table. Your teacher may show videos or give a presentation using a projector.

There are many ways you can learn in the classroom. E-learning is no different. Many different types of media are used for learning. When different types of media, such as video and text, are used together, it is called **multimedia**.

Here is some of the media used in e-learning.

- **Video and interactive video**

  Video can be used for demonstrations, for example, how to use a piece of software. Or it can be used to show what an animal is like, while a commentary or words on-screen give facts about the animal.

  Sometimes you can decide how a video is played. For example, a video clip might be followed by a question. The next clip you see depends on what your answer to the question is. What you see depends on the action you take. This is called **interactive video**.

- **Presentations and audio**

  You have created presentations in earlier books. Some e-learning lessons are shown as a presentation. Presentations show information on a series of slides. An audio file is sometimes used to explain the content on each slide.

- **Text and images**

  Some e-learning material is made up of simple text with images, like the books in your classroom. Text and image content can be cheap and easy to make. It can be created quite quickly by teachers who do not have the IT skills to make videos, games and simulations. Books that you read on-screen are called e-books.

- **Games and simulations**

  You must complete learning challenges to progress in an e-learning game. Games can make learning fun.

  As we saw in Lesson 2.1, a simulation is a model of a real-life situation, for example, a scientific experiment. In a simulation of a chemistry experiment, you can safely see what happens if you increase the temperature or change the number or amount of chemicals.



Some students learn best if they watch a video. Others prefer to read, or to use a simulation. E-learning uses a mix of media types. That allows students to learn using the materials that suit them best.



Activity

How do you prefer to learn things? Do you prefer to read or watch a video? Do you think an educational game would help you learn?

How important is it for your teacher to explain something to you?

Tests and assignments

When you study at school, you do tests and exams. You do assignments and complete homework.

There are also tests and assignments in e-learning.

Tests are made up of short questions. For each question, you choose an answer from a list of options, or provide a single-word answer. This type of question can be marked automatically by a computer. This saves time and provides you with instant results.

Assignments are longer pieces of work that a trainer or teacher must mark. You send the assignment using the computer and get your mark back in the same way.

How is e-learning taught?

Technology is changing the way that people work. Trainers and teachers are still important in e-learning but their job is not the same as in classroom teaching.

In e-learning, teachers and trainers spend less time:

- teaching classes of students
- marking work
- working face to face with students.



This gives them more time to:

- help students get better results
- talk to students online and answer messages
- help individual students
- create e-learning media.

More learning is delivered on-screen, so a trainer or teacher does not spend as much time teaching in front of a full class. This gives them more time to spend helping individual students.

Advantages and disadvantages of e-learning



Advantages

Here are some of the advantages of e-learning.

- You can learn where and when you want to.
- If you live in a remote area, you can learn without having to travel long distances.
- You can use the type of media you prefer.
- Teachers have more time to help individual students.

Disadvantages

And here are a couple of disadvantages.

- Some people prefer to learn in a class with other people.
- It can be difficult to work without a strict timetable.

Digital citizen of the future

Technology is changing quickly. This means you will need to learn new skills throughout your working life. E-learning is one way for you to keep your skills and knowledge up to date.

Creating e-learning materials is a job that combines creative work with computing skills. Is it a career you would consider in the future?

Activity

Plan a lesson for a subject you are interested in.

- State the topic of the lesson.
- Outline the information you will include in your lesson.
- Write down one or two questions you could ask to see if the learner has understood the multimedia content.

Extra challenge

Do online research to find an example of multimedia content you could use for the lesson you planned.

Test

1. Explain what multimedia means in the context of e-learning.
2. Give three examples of media types that can be used for e-learning.
3. List three ways that a teacher's job will change if e-learning is used in your school.
4. Describe three ways that e-learning could help you to learn.

2.4 Starting your e-learning project

In this lesson

You will learn:

- how to plan your e-learning project.

Planning your e-learning project

For the rest of this unit, you will work in small groups to create a short e-learning course. There are some suggested topics below, but you can also choose your own topic.

Each member in the group will produce:

- a slide that covers one lesson
- a test question.

You will make the e-learning content using presentation software. The examples in this unit use Microsoft PowerPoint. If you use different presentation software, your teacher will tell you the differences between your software and PowerPoint.



First, plan how your team will work together. Here are some things you need to think about.

Choose a topic

It's up to your team to choose a topic. You could choose something you have studied in your computing course. Or choose a topic from another school subject. Here are some computing ideas.

- Network hardware (Unit 1 of this book)
- Computers and space discovery (see Lesson 2.2)
- Computer hardware (input, output and storage devices)
- Cybercrime (Unit 2, Student Book 7)
- Staying safe on the internet (Unit 2, Student Book 7).

Agree who will do what

When you have chosen a topic, decide how you will split it up into lessons. You need one lesson for each member of the group. For example, if you choose computer output devices, you could decide to divide your course like this:

- What is an output device?
- Computer screens
- Printers
- Speakers.

You need one heading for each member of your team.

Design your presentation

When you have completed your e-learning lesson, you will put it together with the lessons made by other members of your team. The completed presentation needs to look good. Each lesson needs to have the same style so that it looks like one person has written the whole course.

Here are some ways you can help ensure you have a consistent style.

- Agree a **theme** to use for your completed course. Your presentation software will have backgrounds you can choose from.
- Agree a **template** to use for your content. Your presentation software will have templates you can use. A layout like the one below will be suitable. There is room for a heading and an image with text next to it.

The themes that you can apply to your presentation are in the 'Design' tab. Choose one with a white background. It will make your text easier to read.

You can see what page templates are available if you click the 'Layout' button in the 'Slides' section in the 'Home' tab.

Your teacher will show you an example of a completed lesson.



Starting your project

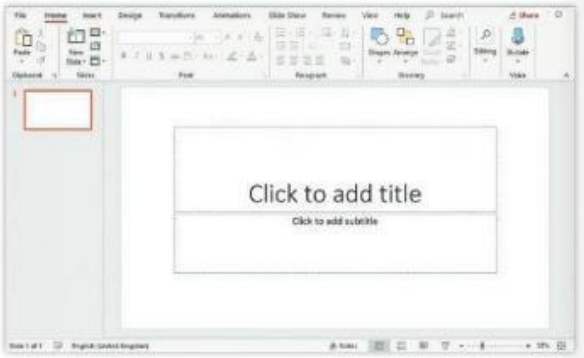
In Lesson 2.5, you will create your individual e-learning slides.

First, you will complete the title slide and contents page for your e-learning course.

Add a title slide

To add a title slide, open a new, blank presentation in your presentation software.

A blank title slide should be automatically inserted. If not, add a title slide.

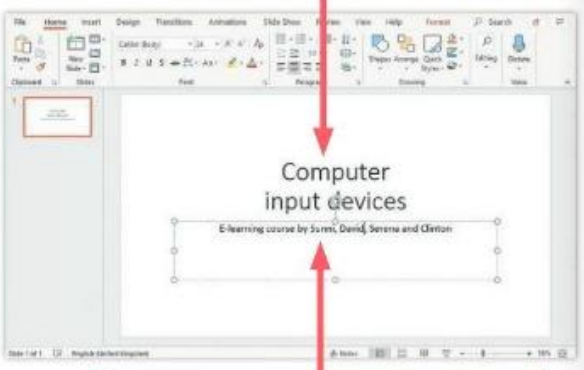


Add a title and credits

Now you are going to add a title and credits to your e-learning course.

Agree a suitable title for your presentation. It needs to describe what your e-learning course is about. It must not be too long.

1 Add a title to your presentation.



2 Add the names of your team members.

Choose a theme

Adding a theme to your presentation will make it look more professional. The theme you choose will appear on every slide, so try a few background designs before you make your final choice. Pick a theme that leaves plenty of space in the centre of your slides. A white background will make your text easier to read.

1 Select the 'Design' tab in your presentation software.

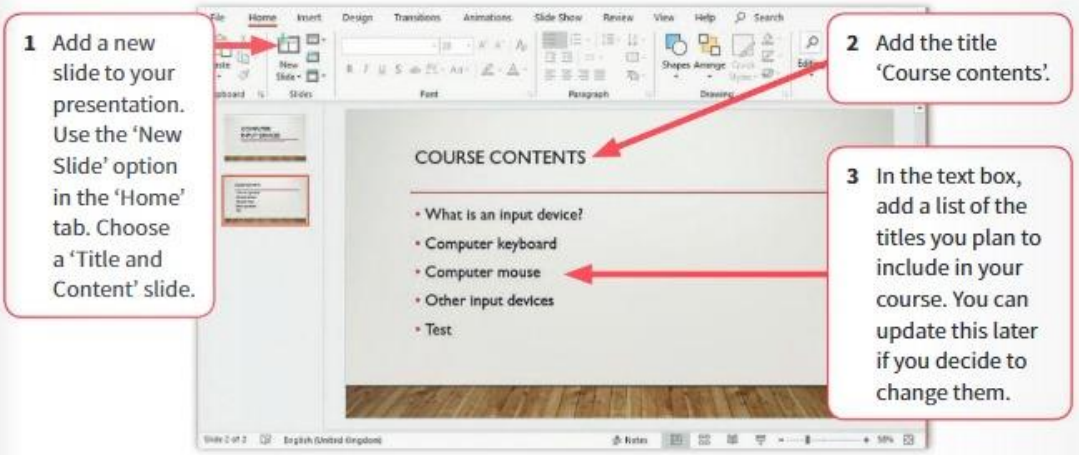
2 Browse the 'Themes' menu and select a theme.



Add a contents slide

Create a contents slide for your e-learning course. A contents page tells students what they will learn about when they study a course.

The contents slide should list the titles of the lessons your team members are going to create.



Save your work. You will use the presentation again in the next lesson.

When each member of your team has finished their slides, you will need to bring them all into a single presentation. Discuss how your team will do that.

Activity

Create the title and content slides for the e-learning course you have planned.



Extra challenge

Add an image to the title page. Choose an image that is relevant to your presentation. You can select an image using the 'Online Pictures' option in the 'Insert' tab.

Test

1. Why does your e-learning course need a title slide?
2. How do you add a new slide to a presentation?
3. Give reasons why you chose the theme for your presentation.
4. How will you carry out research for your e-learning lesson?